Practice workbook · Ethics of technology: privacy, inclusion and professionalism

Barriers to equal access, reluctance to use online services, privacy concerns of surveillance, and how computing professionals show ethical behaviour.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-15

1. Question 4 · part f · 6 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part f only.

4 Peter and Marie are employees at a research facility.

(a) Peter is creating a predator-prey population simulation of rabbits and foxes.

He creates rules to simulate the movement and feeding patterns for each animal over time.

One benefit of the predator-prey simulation is that it is simpler than monitoring real life animal populations.

Explain **one other** benefit of using this simulation.

(2)

(b) Identify the computational model in which several computer systems work together to achieve a goal.

(1)

- ☒ **A** Sequential model
- ☒ **B** Parallel model
- ☒ **C** Multi-agent model
- ☒ **D** Input-process-output model

(c) Employees at the research facility use a biometric identifier to login to the network.

Describe **one other** security measure that could be used to ensure that an employee is who they claim to be.

(2)

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1. Question 4 · part f · 6 marks

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Worked-example retry · Attempt part f only.

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(d) Marie is a researcher in nanotechnology.

(i) State what is meant by the term **nanotechnology**.

(1)

(ii) Give **one** application of nanotechnology.

(1)

(e) Marie has a mobile phone.

The address of the device is:

2024:0eb8:3c4d:0015:0000:0000:1a2f:1a2b

Describe the relationship between the structure of this address and the total number of bits required to store it.

(3)

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1. Question 4 · part f · 6 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part f only.

(f) The research facility is considering introducing cameras and facial recognition software to monitor the work and movement of people on and around its site.

Discuss the privacy issues that need to be considered when deciding whether or not to install this new system.

(6)

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(Total for Question 4 = 16 marks)

1. Question 4 · part f · 6 marks · Working and answer

Working and answer

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2. Question 6 · part b · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

6 (a) Two emerging technologies are DNA computing and quantum computing.

(i) Explain **one** difference between DNA computing and traditional computing.

(2)

(ii) Give a definition of the term **superposition** in quantum computing.

(1)

(b) Ethically, everyone should be able to access the benefits of technology.

Give **two** reasons why this may not be possible.

(2)

1

2

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16

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2. Question 6 · part b · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b only.

(c) Widespread use of computer technology has a negative environmental impact.

Discuss the negative environmental impact of using computers and actions that could be taken to reduce it.

(6)

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(Total for Question 6 = 11 marks)

2. Question 6 · part b · 2 marks · Working and answer

Working and answer

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3. Question 5 · part a.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part a.ii only.

5 Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

(2)

(ii) Give **one** reason why a patient may not want to use this online service.

(1)

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

(1)

(ii) Name the type of network used to access the server from the Stockton building.

(1)

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3. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

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4. Question 1 · part e · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part e only.

1 People use networks in all aspects of their lives.

(a) An office building has desktop computers connected in a network.

One reason for networking computers is to enable technicians to maintain them remotely.

State **two other** reasons for connecting computers in a network.

(2)

1 .....

.....

2 .....

.....

(b) Access to data must be controlled.

(i) Identify **one** reason that encryption is used.

(1)

☒ **A** To compress the file so it takes up less storage space

☒ **B** To make sure both sender and receiver use the same key

☒ **C** To make sure data is only understood by the intended receiver

☒ **D** To stop malicious hackers getting into a computer network

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4. Question 1 · part e · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part e only.

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(ii) File servers are in a small room at the back of an office.

The office has a burglar alarm.

Give **two other** ways that the servers can be secured using physical methods.

(2)

1 .....

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2 .....

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(iii) Phishing is a type of social engineering.

State what is meant by the term **social engineering**.

(1)

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(c) 5G is a communication standard.

Identify the characteristic that is true for the 5G communication standard.

(1)

- ☒ **A** It has a high transmission latency
- ☒ **B** It has built-in security
- ☒ **C** It has a lower bandwidth than 3G and 4G
- ☒ **D** It uses wireless communication

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4. Question 1 · part e · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part e only.

(d) Programmers write and test code.

They use modular testing.

State what is meant by the term **modular testing**.

(1)

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(e) Computer scientists are viewed as professionals, in the same way that doctors are viewed as professionals.

Give **two** ways that computer scientists can demonstrate professionalism.

(2)

1 .....

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2 .....

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(Total for Question 1 = 10 marks)

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Working and answer

5. Question 5 · part b.iii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.iii only.

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5 Smartwatches are a form of wearable technology.

(a) Embedded systems are used in smartwatches.

(i) A smartwatch has a step tracker that uses a built-in accelerometer sensor.

Complete the table by adding **one other** built-in sensor a smartwatch could use and its purpose.

(2)

Sensor	Purpose
Accelerometer	To measure body movement and step count

(ii) One characteristic of an embedded system in a smartwatch is that it uses sensors.

Give **two other** characteristics of an embedded system that make it suitable for use in a smartwatch.

(2)

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1 .....

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2 .....

.....

(b) A software development team writes the software for a smartwatch using assembly language.

(i) Identify the purpose of an assembler.

(1)

- ☒ **A** To convert a low-level language into a high-level language
- ☒ **B** To convert low-level source code into machine code
- ☒ **C** To execute machine code
- ☒ **D** To translate a high-level language into a low-level language

5. Question 5 · part b.iii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.iii only.

(ii) Explain **one** benefit of writing the software for the smartwatch in a low-level language.

(2)

(iii) Members of the software development team are computing professionals.

Give **two** ways that members of the team can demonstrate ethical behaviour.

(2)

1 .....

2 .....

(iv) The software for the smartwatch takes up 500,000 bytes of storage.

Identify the value that is equivalent to 500,000 bytes.

(1)

- ☒ **A** 0.5 MB
- ☒ **B** 500 KiB
- ☒ **C** 0.5 MiB
- ☒ **D** 5 MB

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P 7 8 6 3 7 A 0 1 4 2 0

5. Question 5 · part b.iii · 2 marks · Working and answer

Working and answer

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